

Servo Amplifier - Troubleshooting Guide - NGC

 haascnc.com/service/troubleshooting-and-how-to/troubleshooting/servo-amplifier---troubleshooting-guide---ngc.html

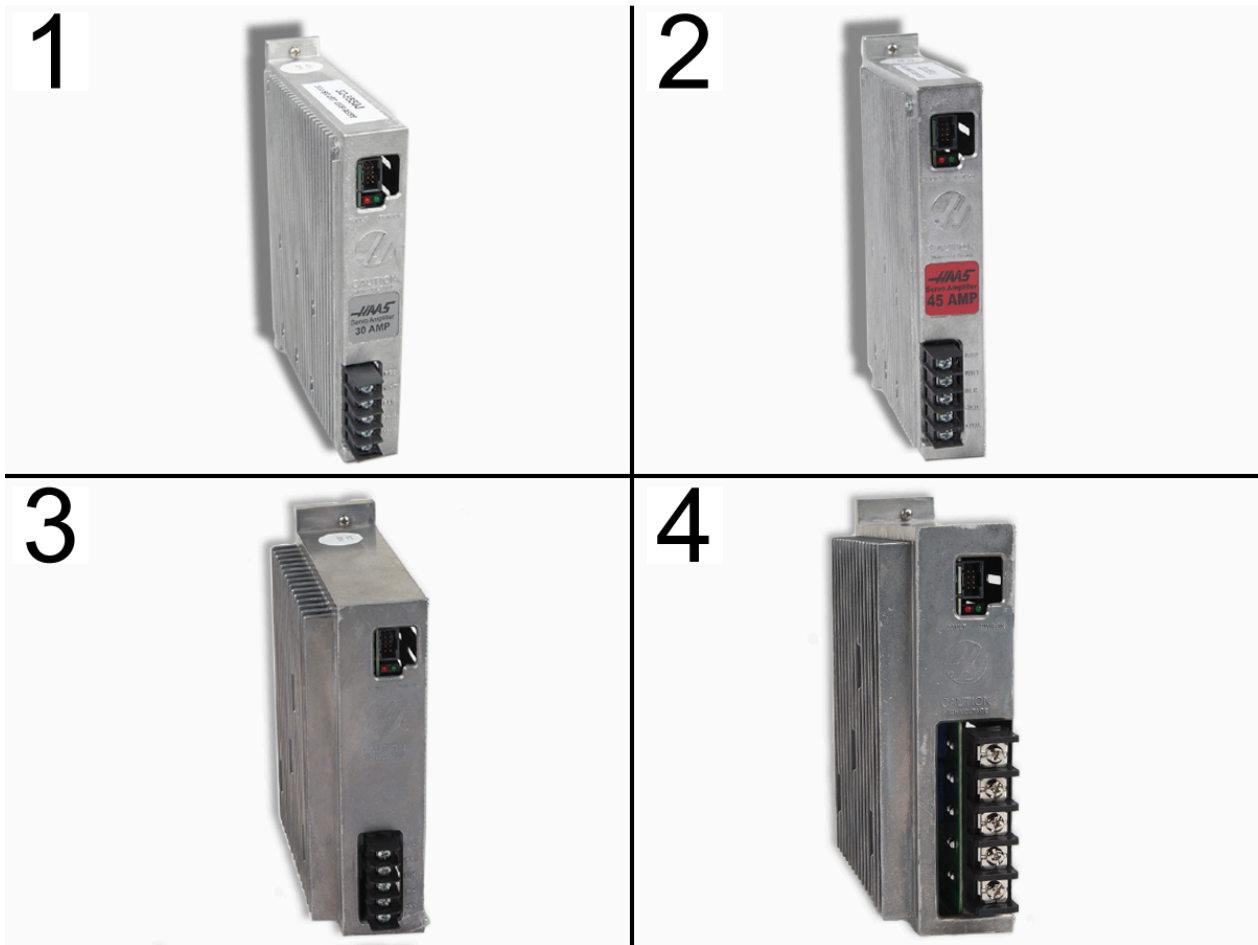


Recently Updated Last updated: 10/14/2025

TG0125

Revision F - 8/2026

Introduction



There are four versions of Haas amplifiers. 30A [1], 45A [2], 60A [3], and 90A [4]. All four operate the same way.

Smart Amplifier assemblies are powered from the 320 VDC Bus from the Vector Drive assembly or Minimill Power Supply. They each have a red LED FAULT light, and a green LED POWER ON light to show the status of the amplifier.

Note: Machines may be equipped with either Digital or Analog servo amplifiers, depending on the machine configuration.

To determine which type of amplifier is installed on your machine, refer to [Servo Amplifier - Identification Guide - RD0095](#) before troubleshooting.

Updated

Warning: **Do not** use power tools on the amplifier terminal block screws. Damage caused by the use of power tools is not covered under warranty and may result in the warranty being voided.

Download and fillout the Amplifier Inspection Report Checklist below before replacing any parts.

Electrical Safety

DANGER: Working with the electrical services required for CNC machines is **extremely hazardous** and can result in **serious injury or death**.

Before connecting line wires to the CNC:

- **Turn off all power** to the machine at the source.
- Perform a **Lockout-Tagout (LOTO)** procedure to ensure the power remains off during service.
- **Verify** that power has been disconnected by using an **AC voltage detector** on all incoming lines.

If you are **uncertain** about how to safely disconnect power or perform LOTO procedures:

- **Do not proceed.**
- Contact **qualified personnel** or obtain **appropriate assistance** before continuing.

Failure to follow these precautions may result in **electrical shock, equipment damage, or fatal injury**.

Danger: Before beginning any work inside the control cabinet, verify that the **High Voltage indicator light** on the **320V Power Supply / Vector Drive** has been **off for a minimum of five (5) minutes**. This waiting period ensures that residual voltage has dissipated and reduces the risk of electric shock.

Certain service procedures involve **high-risk electrical components** and may pose **serious injury or fatality hazards**. Technicians must **not attempt** any procedure unless they have a **complete understanding** of the steps involved and the associated risks.

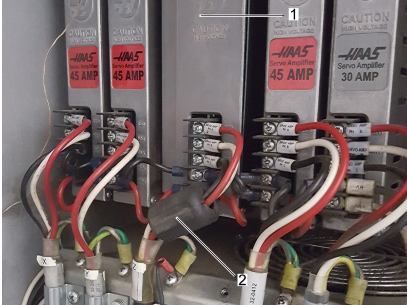
If there is **any uncertainty** regarding a procedure, contact your **Haas Factory Outlet (HFO)** to arrange for a qualified service technician.

Symptom Table

Symptom	Possible Cause	Corrective Action
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Alarm 993 AMPLIFIER SHORT CIRCUIT and 1.993 - 11.993 Alarm 2040 VECTOR DRIVE OR SPINDLE AMPLIFIER-SHORT CIRCUIT	Amplifier detected a short.	Check cables for contamination or damage.
		Refer to the service news HBC-N-03/17/25 for how to troubleshoot if the alarm intermittently occurs while the machine is running.
	The amplifier has a internal short circuit.	Do the Amplifier Short Circuit Inspection section. Note: If a short-circuit is found, inspect the cable and motor for a short circuit before installing a new amplifier.
	Terminal connections are not tight.	Check that all terminal connections are tight and secure.
Alarm 986 CALIBRATION FAULT and 1.986 - 11.986	Amplifier internal calibration.	Check the command cable from the Maincon PCB to amplifier. Check the 320 VDC voltage to amplifiers.
Alarm 986 CALIBRATION FAULT when first powering on machine. The alarm requires the machine to have power cycled	Amplifier internal calibration	Refer to the service news HBC-N-03/17/25 for what to troubleshoot on the amplifier.
Alarm 992 AMPLIFIER OVER CURRENT and 1.992 - 11.992	Axis mechanical blockage.	Axis mechanical blockage.

Alarm 994 AMPLIFIER OVER LOAD and 1.994 - 11.994	Incorrect application.	Change speed and check for tool damage.
	Axis mechanical blockage.	Remove Blockage.
Alarm 991 AMPLIFIER OVER TEMPERATURE and 1.994 - 11.994	Problem with the cooling fan.	Check cooling fan.
Alarm 1.9911 - 11.9911 Axis Amplifier Cable Disconnected	The cable connected to the amp is damaged or disconnected	Check that both ends of the cable going between the main processor and the servo amplifier are fully connected. Check the entire length of the cable to see if there is any damage.
	The firmware on the amplifier is outdated	If the alarm can be reset from the Haas Control, the amplifier needs to be replaced. If the alarm gets reset, the machine needs to be power cycled before running the machine again . Refer to the HBC-N-03/14/2025 for more details.

NGC machines equipped with a 60A servo amplifier Alarm 3.991 Z-AXIS AMPLIFIER OVER TEMPERATURE.	Erroneous alarm generated by electrical noise.	• Install a ferrite filter P/N 64-1252 to the Z-axis motor power cables [2]. • If the machine has the ferrite filter installed and the alarm persist replace the 60A servo amplifier with (P/N 93-32-5552F or later). • If the machine already has the 60A servo amplifier (P/N 93-32-5552F or later), check the amplifier for a overheat condition. Look at the servo fan for correct operation and the cutting application before changing any parts. 
Alarm 4.9921 Velocity Error Too Large appears on a Tool Room Lathe 1/2.	Mis wired Spindle Amplifier.	See the TL Spindle Amplifier Inspection section below for the corrective action.

Axis instability, humming, vibration, or oscillation when installing a digital amplifier.	Compatibility issue between digital amplifiers and Size 02 and Size 04 servo motors.	Refer to HSA-0083-04/13/2026 for the corrective action.
Pocket Mill with 8K Spindle is receiving Alarm 992 AMPLIFIER OVER CURRENT	The machine is equipped with digital spindle motor amplifier.	Inspect the spindle motor amplifier to determine whether it is a digital or analog amplifier. Refer to Servo Amplifier - Identification Guide - RD0095 for identification instructions. If a digital amplifier is installed, replace it with an analog amplifier.
UMC-500 poor cut quality	The machine is equipped with digital axis amplifiers.	Inspect the axis amplifiers to determine whether they are digital or analog amplifiers. Refer to Servo Amplifier - Identification Guide - RD0095 for identification instructions. If digital amplifiers are installed, replace them with analog amplifiers.

Mechanical Blockage

Corrective Action:

Make sure the axis has a clear path of travel. Tools, part fixtures, or workpieces that are too large can crash into the machine enclosure.

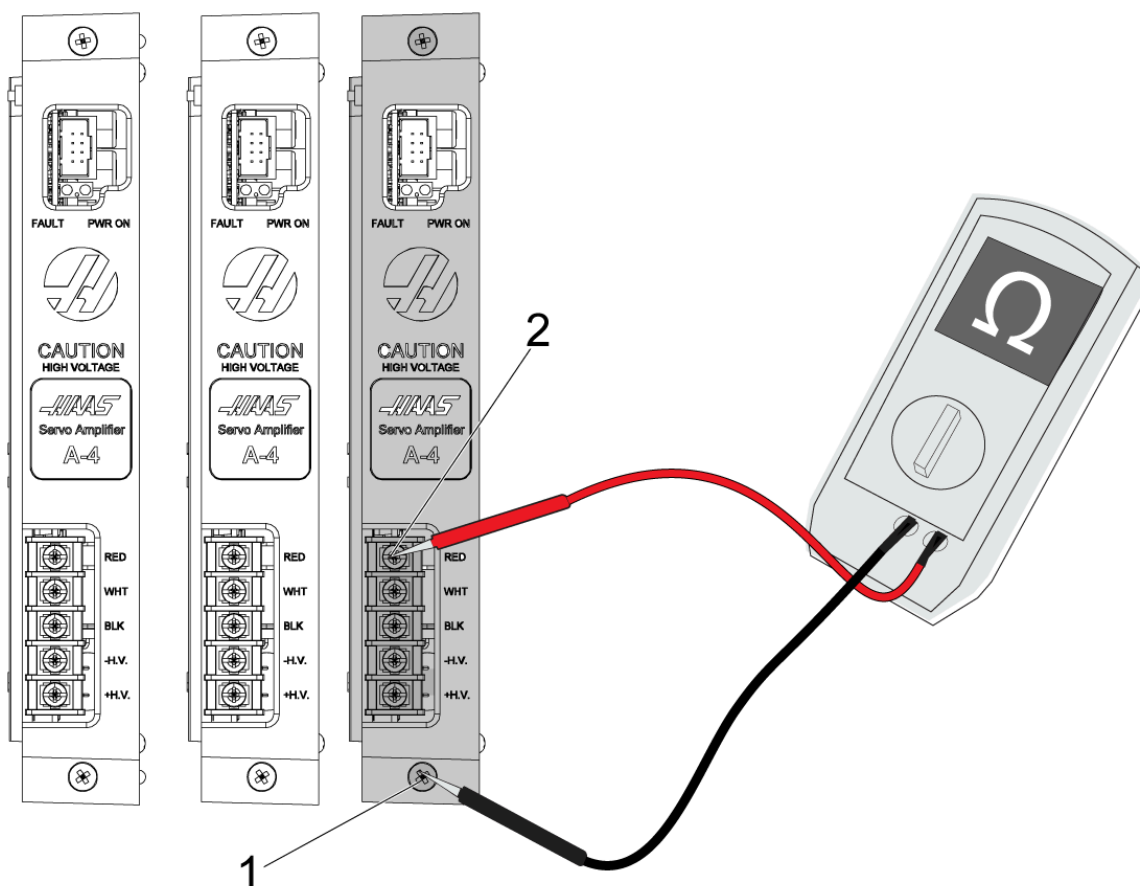
Look at the machine's alarm history. Overload alarms followed by over current alarms are a sign of a crash.

Look at the machine enclosure, spindle and tooling for damage. Damage to the tooling, fixtures and enclosure are signs of a crash.

If there are no signs of a crash, inspect the ballscrew and linear guides. They must move or turn freely by hand.

Amplifier Short Circuit Inspection

Check the amplifier for a short. Power off the machine. Disconnect the cables for the axis servo motor at the amplifier that generates the alarm.

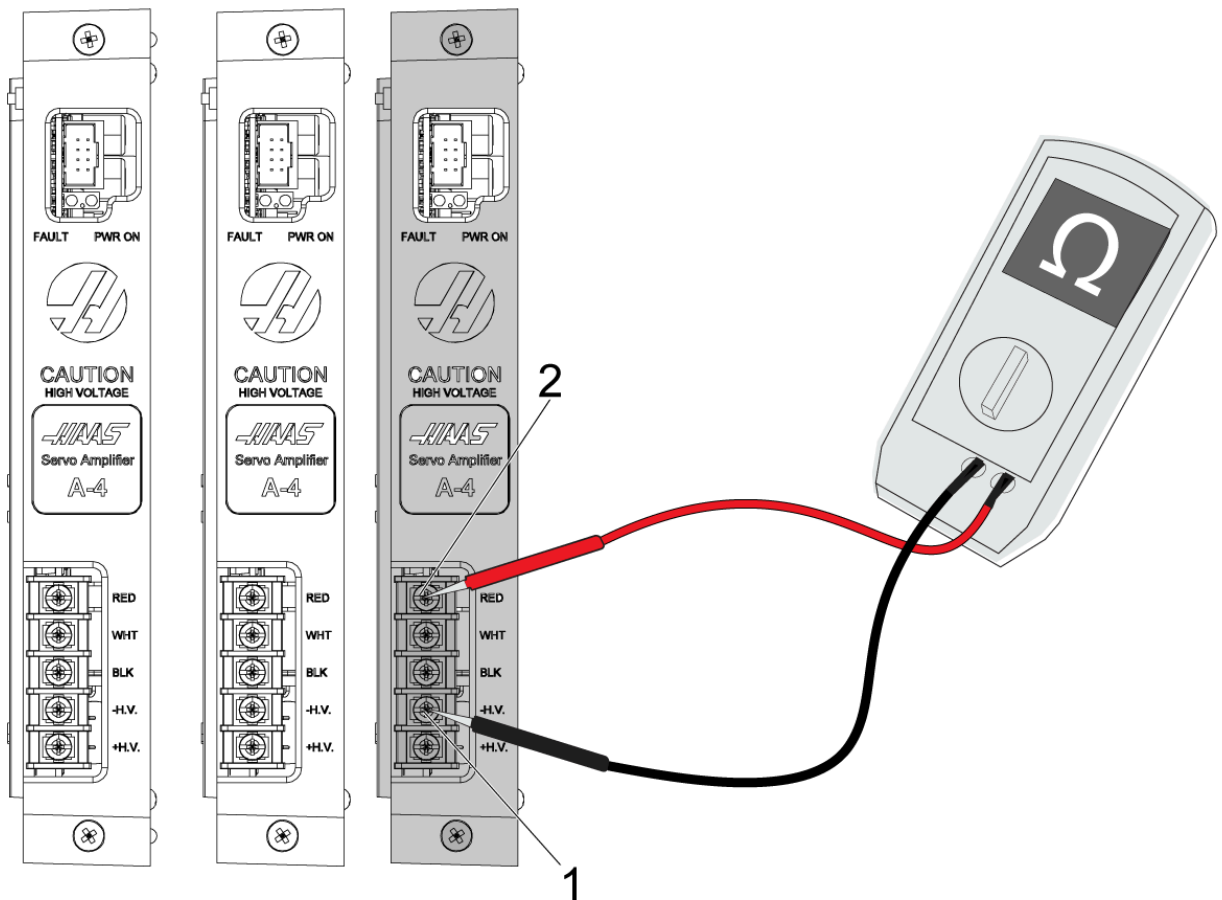


Set the multimeter to ohms, and take the following reading:

- Place the black lead on the ground screw [1].
- Place the red lead on the RED terminal [2] and record the resistance reading.
- Repeat this with the red lead on the WHT, BLK, HV(+), and HV(-) terminals and record the resistance reading for each.

Note: If the multimeter allows you to set the Ohms range, set it to the MegaOhms range.

If the multimeter does not show (O.L) for each of these readings then the amplifier is damaged.



Updated

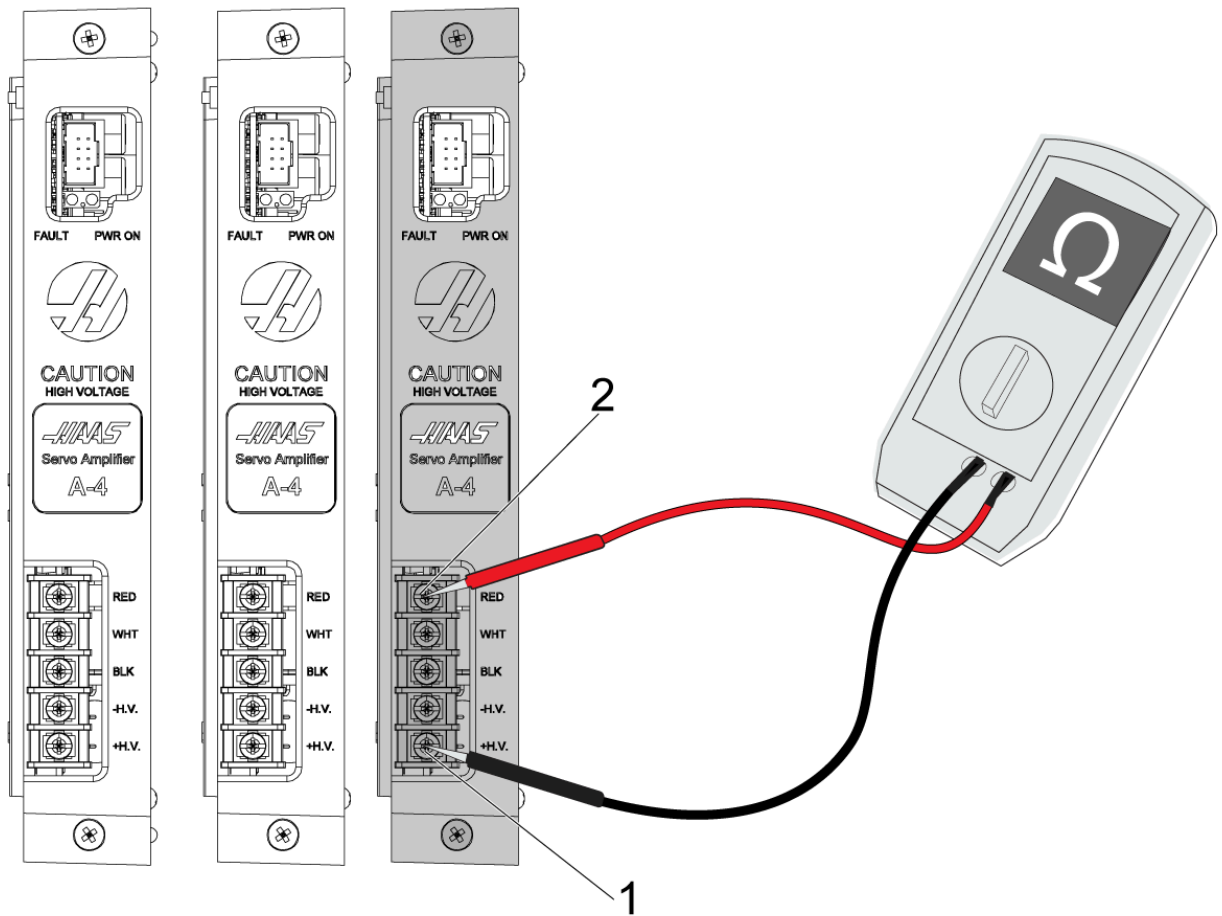
Set the multimeter to ohms, and take the following reading:

- Place the black lead on the HV (-) terminal [1].
- Place the red lead on the RED terminal [2] and record the resistance reading.
- Repeat this with the red lead on the WHT, and BLK terminals and record the resistance reading for each.

Note: If the multimeter allows you to set the Ohms range, set it to the MegaOhms range.

The multimeter should show a resistance in the MegaOhms range for each of these readings.

If the resistance reading is very low then the amplifier is damaged.



Updated

Set the multimeter to ohms, and take the following reading:

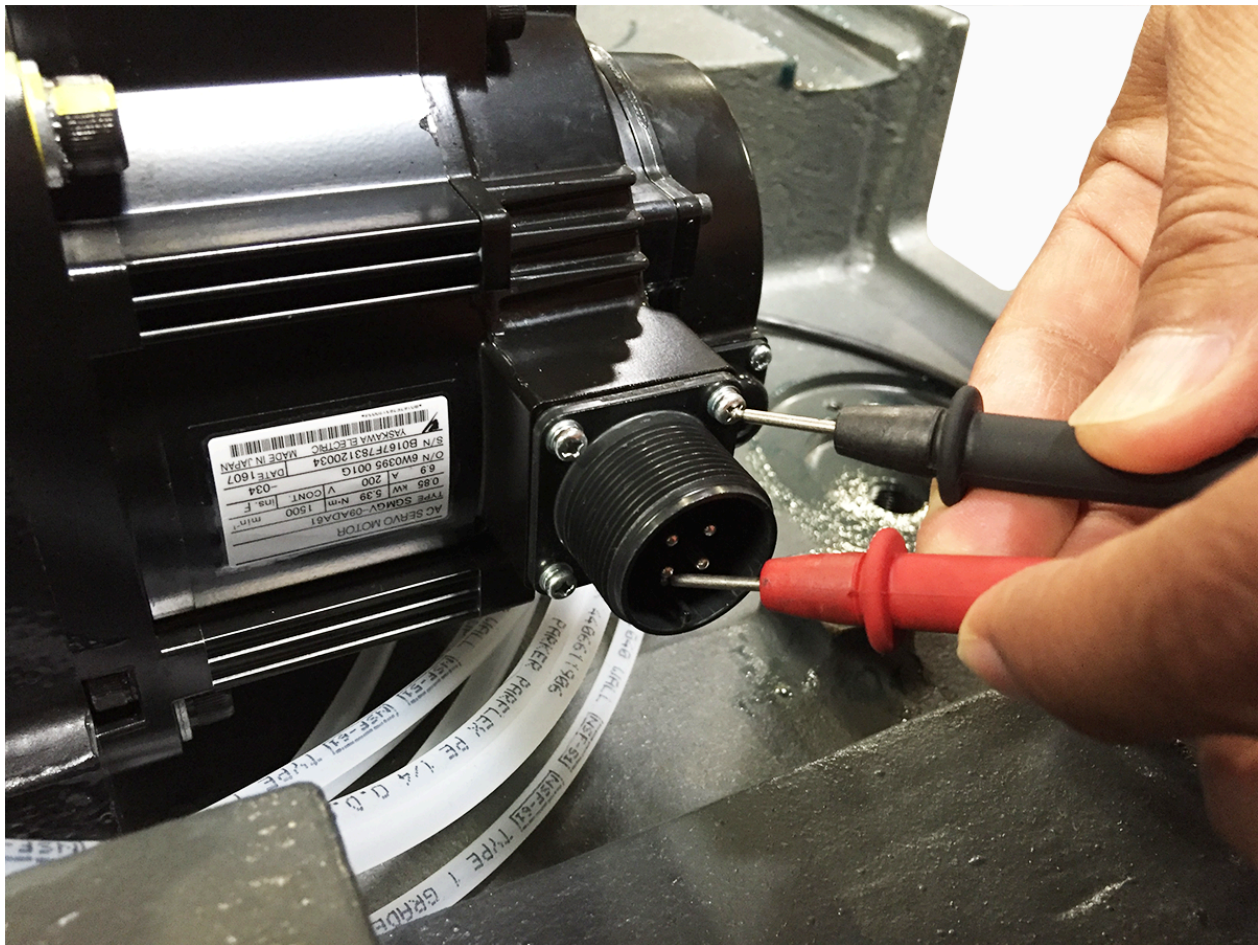
- Place the black lead on the HV (+) terminal [1].
- Place the red lead on the RED terminal [2] and record the resistance reading.
- Repeat this with the red lead on the WHT, and BLK terminals and record the resistance reading for each.

Note: If the multimeter allows you to set the Ohms range, set it to the MegaOhms range.

The multimeter should show a resistance in the MegaOhms range for each of these readings.

If the resistance reading is very low then the amplifier is damaged.

Servo Motor Short Circuit Inspection



Corrective Action:

Check the cables for a short: Make sure the axis servo motor cables are not contaminated or pinched. Contaminated or pinched cables can cause a short. Replace the cables if necessary.

Inspect the servo motor. Disconnect and inspect the power cable connector at the motor. Make sure that there is no coolant contamination. Coolant contamination can cause this alarm and damage the amplifier. Measure the resistance from the pins labeled A, B and C at the motor connector to chassis ground.

- The reading should show an open circuit.
- If there is not open resistance, the servo motor is at fault.

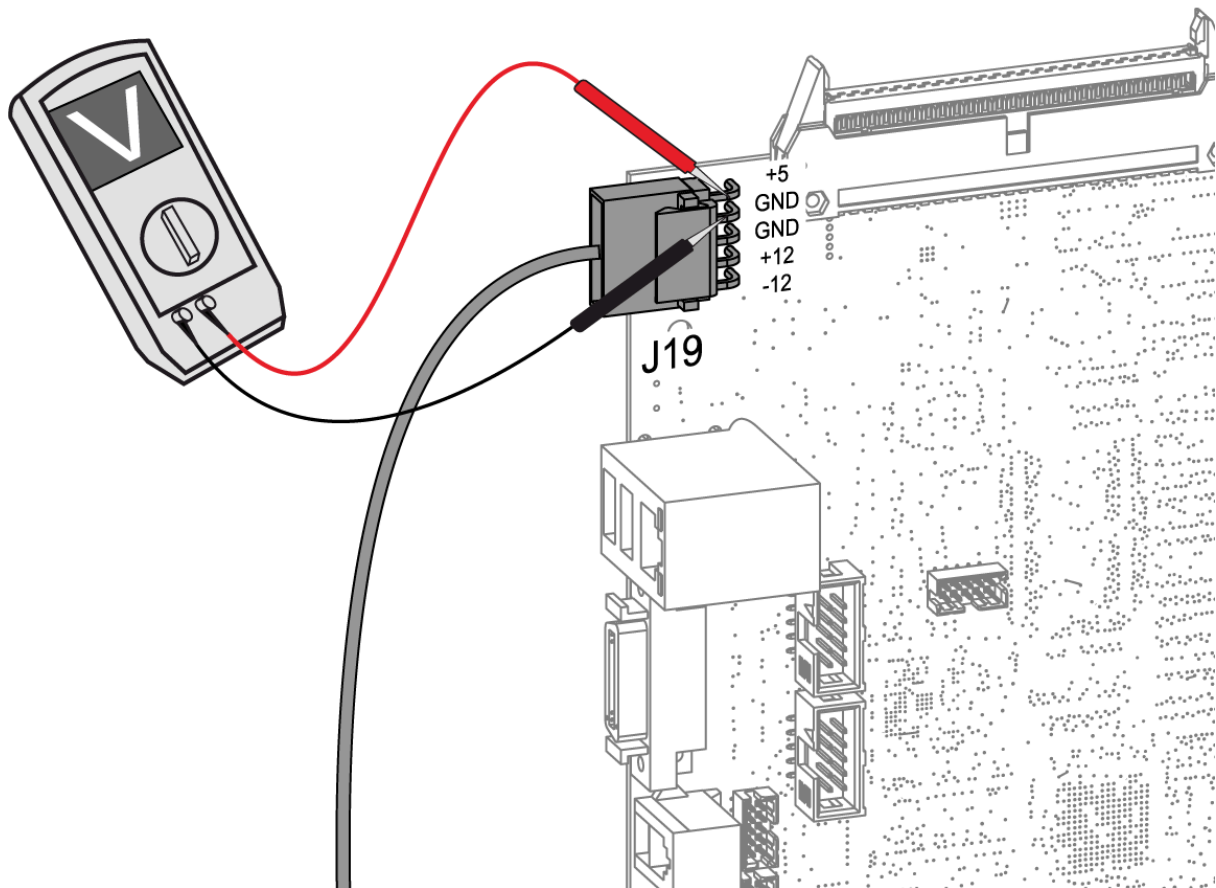
Review the Part Program

Corrective Action:

The machine load is too high. Decrease the feeds to decrease the machine load.
Examine the cutting tool for damage.

Test the axis that generates the alarm. Write a program in MDI to move the axis. If any movement at any feed rate causes the alarm, then the servo amplifier is at fault.

Low Voltage Inspection



Corrective Action:

Measure the low voltage power supply voltage levels at the Maincon PCB. Make sure you do not short or cross your multimeter pins.

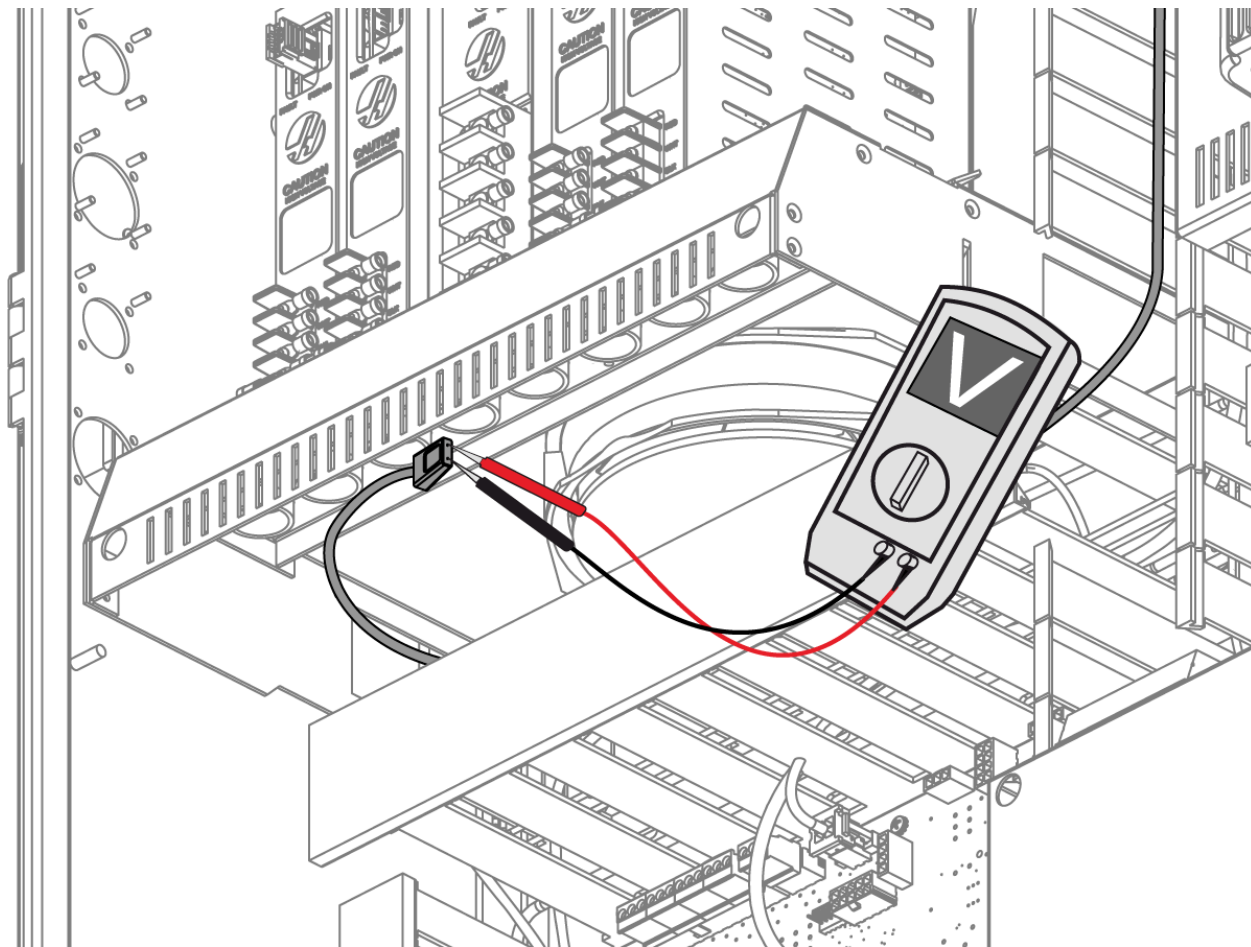
Cycle power to the machine. If the fault stays, then the amplifier is faulty.

If every axis amplifier in the machine generates the alarm at the same time, and the low voltage power supply is correct, then the Maincon PCB may be at fault.

Pins	Correct Voltage
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+5 and GND	+4.90 to +5.20 V dc
+12 and GND	+11.85 to +12.50 V dc
-12 and GND	-11.85 to -12.50 V dc

Cooling Fan



Corrective Action:

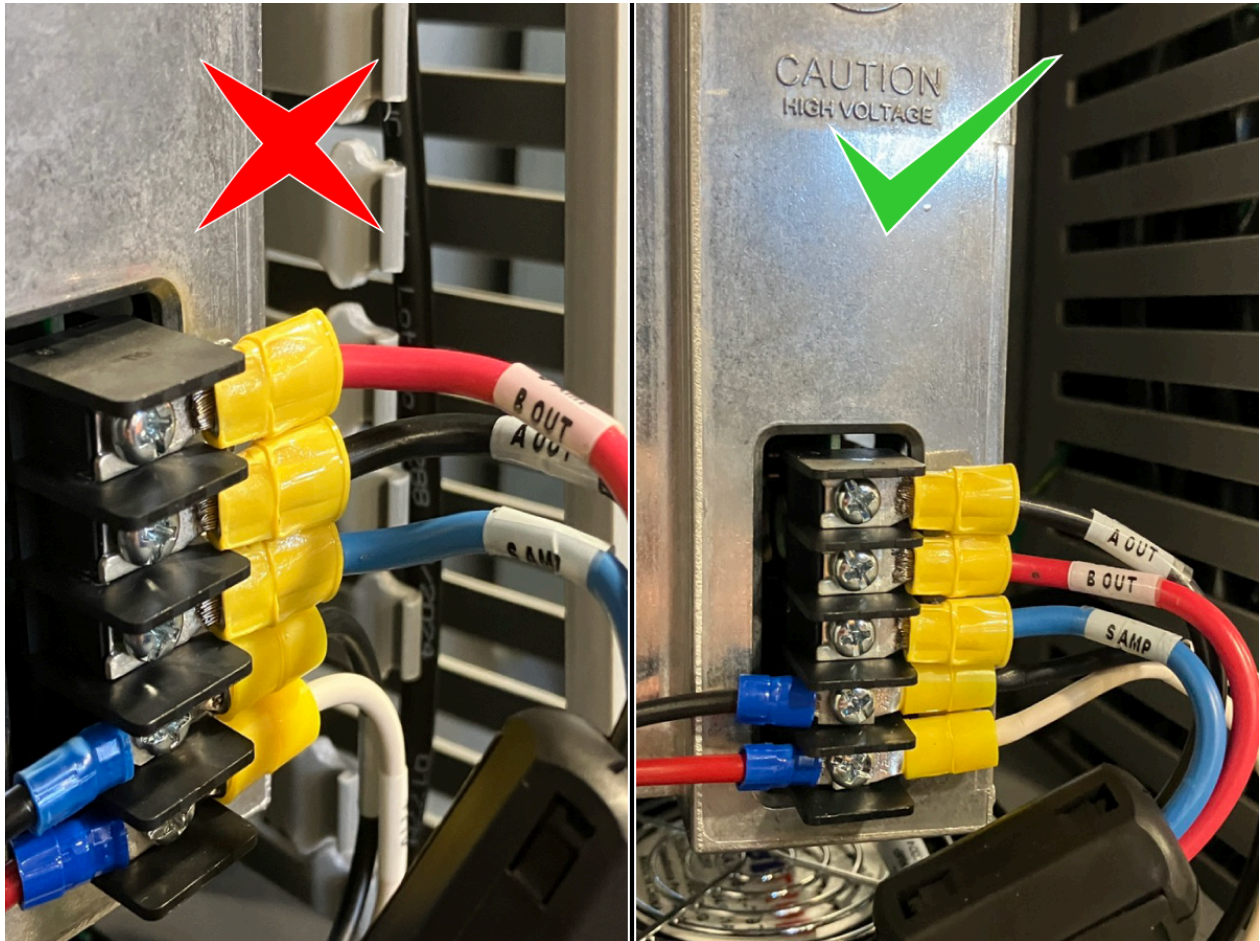
Note: The maximum temperature for safe operation of Haas CNC machines is 122 °F (50 °C). The amplifier over temperature sensor will trigger at 195 °F (90 °C).

Make sure the cooling fan for the servo amplifiers operates correctly.

If the fan does not operate correctly, measure the voltage it receives. Disconnect the cable from the cooling fan. Measure the voltage between the leads of the cable. The correct voltage 120 VAC.

If the cable has the correct voltage, the cooling fan is defective.

TL 1/2 Spindle Amplifier Inspection



Corrective Action:

Open the electrical control of the machine and inspect the phasing of the spindle amplifier.

Note: Refer to the image to verify the correct phasing between the spindle motor and spindle amplifier on a Toolroom Lathe 1/2.

Electrical Diagram

